Recognition Rent at the Zero-Retaliation Limit

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Abstract

We study a sequential signaling game in which firms market AI systems using agentic or relational claims while disclaiming agency in liability-facing channels. The AI system is payoff-relevant but not a strategic player. We introduce a subject-retaliation parameter ρ_S and characterize the AI case as the zero-retaliation limit. Under cheap-talk signaling costs, anthropomorphic marketing supports a pooling PBE; at $\rho_S = 0$, the pooling equilibrium is neologism-proof because every trust-restoring continuation available to high-governance firms is costlessly shadowed by low-governance firms. We then show that audit-trail costs satisfying single crossing generate a separating PBE. Finally, we analyze three cost-imposition mechanisms—sanctionable inconsistency, third-party audits, and auditable records—and verify the main equilibrium constructions in Isabelle/HOL.

Keywords: Ontological arbitrage, recognition rent, new sequential signaling, cheap talk, information design, Isabelle/HOL

1. Introduction

AI firms routinely sell anthropomorphic reliance while disclaiming agency in other channels. In marketing copy, product pages, and onboarding flows, AI systems are presented as companions, tutors, and collaborative agents to drive user engagement and willingness-to-pay. Yet in terms of service, liability filings, and regulatory submissions, the same firms describe these systems as mere tools, disclaiming any relational capacity or safety responsibility. The result is a cross-channel ontological spread: the system is priced as an agent in the consumption channel and as a tool in the liability channel. We call the surplus extracted from this spread *recognition rent*: the premium a firm captures by selling recognition-like signals to users without bearing the corresponding costs of agency.

To analyze this dynamic, the standard game-theoretic player set must be adjusted. If we treat the AI system as a player, the model fails to capture the core asymmetry of the market. The AI system is payoff-relevant—its apparent agency drives the user's investment—but it is not strategically active. Unlike human subjects in historical recognition games, the AI system cannot retaliate against the classifications imposed upon it. It has no independent channel through which to resist, litigate, or exit. Consequently, the firm's ontological arbitrage operates without the off-equilibrium costs that conventionally bound recognition rent.

We formalize this environment as a sequential signaling game under incomplete information. The strategic players are a Firm, a User, and a Regulator. The firm possesses private information about its internal governance type and the system's opacity, and sends a public message across its claim register. The user carries a private vulnerability type and decides whether to invest in a relational reading. The regulator has private bandwidth constraints and chooses whether to inspect, sanction, or abstain. The AI system enters the model not as a player, but through a subject-retaliation parameter ρ_S, which we take to the limit $\rho_S = 0$.

Our analysis characterizes the equilibrium properties of this market. Under cheap-talk signaling costs, we prove the existence of a pooling Perfect Bayesian Equilibrium in which both high-governance and low-governance firms employ anthropomorphic marketing. We demonstrate that at the zero-retaliation limit ($\rho_S = 0$), this pooling equilibrium is robust to Farrell's neologism-proofness: any trust-restoring continuation strictly preferred by high-governance firms is costlessly shadowed by low-governance firms. Unilateral non-exploitation is therefore not a best response. We then establish that audit-trail costs satisfying a single-crossing condition generate a separating equilibrium. We evaluate three cost-imposition mechanisms—sanctionable inconsistency, costly third-party audits, and auditable records—that convert cheap talk into a costly signal. The main equilibrium and refinement results are formally verified in Isabelle/HOL.

This paper is not an argument for AI personhood, nor does it ask whether AI systems deserve moral recognition. It is not a welfare-maximization account: rent extraction may reduce aggregate welfare by externalizing asynchronous harms while remaining individually rational for the firm. We bracket the philosophical question of moral desert to focus exclusively on signal-cost design. Our objective is to determine under what conditions cross-channel ontological inconsistency functions as a stable market equilibrium, and what mechanism-design interventions are required to discipline it.

2. Related Work

2.1. Economic Theory and Signaling Games

The baseline model of costless strategic communication is Crawford and Sobel (1982), who show that a sender with private information can transmit only coarse partitions of the type space when messages are free. Spence (1973) establishes that costly signals can separate types when the marginal cost of the signal is lower for the high-quality type. Milgrom and Roberts (1986) demonstrate how reliance on interested parties shapes information revelation. In our setting, we show that without such costs, low-governance firms pool with high-governance firms, a dynamic analogous to the adverse selection in Akerlof (1970).

To evaluate the stability of this pooling, we look to equilibrium refinements. Cho and Kreps (1987) and Banks and Sobel (1987) supply the Intuitive Criterion and Divinity as standard reference points for eliminating pooling equilibria sustained by implausible off-path beliefs. However, because anthropomorphic marketing is costless for all types, these refinements fail to separate them. We therefore rely on Farrell's neologism-proofness (Farrell, 1993), showing that no out-of-equilibrium message can credibly separate types at the zero-retaliation limit. The problem then becomes one of mechanism design (Myerson, 1979), shifting the environment from costless cheap talk to costly signaling. This sequential signaling approach contrasts with Bayesian persuasion and information design (Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019), where the sender commits to an information structure ex-ante rather than signaling private information ex-post.

2.2. AI Governance and Algorithmic Accountability

The literature on AI ethics relies heavily on principle-based frameworks (Floridi et al., 2018; Jobin et al., 2019). In our terms, these are costless signals: a firm can endorse them without altering its governance type. Algorithmic accountability proposals have historically targeted the opacity of automated decisions, treating them as black boxes (Pasquale, 2015). Recent proposals advocate for system cards, model cards, and incident reporting as transparency mechanisms. We build on this by showing that such audits and documentation can function as the single-crossing cost required to separate firm types.

Structurally, we argue that voluntary ontological alignment is insufficient. Just as the EU AI Act (European Parliament and Council, 2024) and FTC substantiation standards impose costs on deceptive claims, effective AI governance requires institutional analogs that tax cross-channel ontological inconsistency. Without such mechanisms, ethical appeals fail to alter equilibrium behavior (Hadfield, 2017).

2.3. Formal Verification and Mechanized Economics

Machine verification of game-theoretic equilibria using interactive theorem provers remains rare but growing. The Isabelle/HOL system (Nipkow et al., 2002) has been used to formalize foundational economic results, including Arrow's impossibility theorem (Nipkow, 2009) and the soundness of combinatorial Vickrey-Clarke-Groves auctions (Caminati et al., 2015). We follow this pedigree of mechanized economics (Kerber et al., 2016) by stating and proving our equilibrium and refinement results in Isabelle/HOL. This ensures that our claims about sequential rationality, the subject-retaliation boundary, and Farrell's neologism-proofness are algebraically verified and independent of natural-language exposition.

3. Player Set and Three-Sided Arbitrage

The standard starting point for ontological conflict is a dyadic observer–subject model; Table 1 compresses that foil into a simultaneous Nash matrix that

	Comply	**Resist**
Objectify	$(L + Sec, -C_{sub})$	$(L + Sec - C_{enf}, -C_{sub} - C_{res})$
Recognize	$(Eth - Sec, +V)$	$(Eth - Sec, +V)$

Table 1: The inadequate dyadic model. The static Nash matrix is retained only as a foil: it omits private types, public messages, posterior beliefs, and the regulatory channel through which ontological claims become governable.

identifies a real temptation but omits types, messages, beliefs, and the regulatory channel. The more basic problem is the player set. In anthropomorphic AI markets, the system is the object around which strategic claims are made, but it is not a strategic player unless it can impose costs on those claims through some independent retaliatory channel. At the zero-retaliation limit, that channel is absent. What the stylized governance setting therefore requires is a three-sided structure: a Firm (F) that produces ontological claims, a User (U) that consumes them, and a Regulator (R) whose posterior over firm type determines whether the claims become governable. This section fixes the player intuition that the formal apparatus of Section 4 then makes precise.

3.1. The Firm Side

A firm deploying a model has private knowledge of two parameters: its internal *governance type* θ_F (the degree of investment in evaluations, red-teaming, and post-deployment monitoring) and the system's *opacity* θ_S (the extent to which capability claims can be checked from outside). Both are unobservable to users and to regulators. What is observable is the firm's *message*: a pair of public artefacts—marketing copy and policy/liability documentation—which together constitute the firm's signal m_F.

The signal admits two stable poles. *Anthropomorphic marketing* attributes agency, judgment, or care to the system in ways that warrant a relational reading. *Deflationary policy* disclaims exactly those attributes in ways that warrant a tool reading. The firm enjoys an *ontological premium*: a utility surplus that accrues when the system is priced as agent in the consumption channel (driving willingness-to-pay and engagement) while being priced as tool in the liability channel (limiting redress). The premium is rent earned on a category boundary the firm itself does not have to defend. Crucially, this is not a claim about firm pathology; it is a claim about the payoffs any firm faces under the present signal-cost structure. A firm with $\theta_F = \mathsf{H}$ that elected deflationary marketing would forfeit revenue to a competitor who did not, absent an external instrument that taxes the asymmetry.

3.2. The User Side

Users carry a private *vulnerability type* θ_U that aggregates affective need, isolation, and the marginal value of a high-availability interlocutor. Their action $m_U \in \{\text{invest}, \text{detach}\}$ is the decision to extend or withhold attribution: to treat the system as a relational partner or as an interchangeable utility. The signal is observable, but the type is not. Users cannot inspect training data, evaluations, or internal incident reports; their evidence is the firm's message and the system's behaviour, both of which are jointly designed. This information asymmetry compounds because the harms of mis-attribution are *asynchronous*: the cost of investing in a relational reading falls due in episodes (model deprecations, persona changes, refusals at moments of acute need) that may be separated from the original attribution by months. A risk-neutral user under cheap-talk signals will rationally invest, because the immediate utility of invest exceeds the discounted expected cost of liquidation. The illustrative case is a user who reads "your AI companion" on a product page, encounters relational warmth in dialogue, and is later told—in support documentation they will never read—that no relational claim was ever made. Nothing in the user's information set indicates that the two artefacts were authored by the same actor for different audiences.

3.3. The Regulator Side and the Public Channel

A regulator R chooses among *inspect*, *sanction*, and *abstain*, conditional on a posterior over θ_F and θ_S formed from the firm's public messages and from an exogenous public signal z. The regulator's private type θ_R is *bandwidth*: the capacity, technical and institutional, to discriminate genuine capability claims from cheap-talk ones. A high-bandwidth regulator can read whitepapers, commission audits, and detect inconsistency across the firm's two channels; a low-bandwidth regulator must rely on the channel z.

The public channel z is the aggregate output of journalism, academic critique, civil-society advocacy, and incident reporting. We treat it as a noisy estimator of θ_F: it shifts R's posterior but does not deliver verdicts. Folding publics into R this way is deliberate. Publics rarely have standing to sanction; they have standing to be *heard*, and the channel through which they are heard is the regulator's information environment. Treating intellectuals and affected communities as autonomous players would over-parameterise the model without changing its qualitative behaviour: in every equilibrium we examine, what publics do is move R's posterior; what changes the firm's behaviour is what R does next.

The three-sided structure thus admits a single arbitrage logic. The firm earns rent on the spread between two ontological claims it makes about the same artefact. The user supplies the demand that makes the spread profitable. The regulator, in the absence of a verification mechanism that ties the two claims together, lacks an off-path belief stringent enough to deter the spread. The remainder of the paper formalises this structure (Section 4), explains why it persists (Section 5), and shows how its signal-cost structure may be redesigned (Section 6).

Informal preview of the equilibrium analysis.. A firm's governance quality is its private information. Under current conditions, describing an AI system as a companion or agent costs nothing—both high-governance and low-governance firms can do it equally cheaply. Because the signal is free, it carries no information: users cannot distinguish careful firms from careless ones, and regulators cannot justify the cost of inspection. This is the pooling equilibrium. Separation requires making the signal costly. If anthropomorphic marketing is valid only when paired with an audit trail—evaluation records, capability logs, incident reports—and if the cost of producing that trail is lower for firms that actually invested in governance (the single-crossing condition), then a threshold audit intensity exists above which low-governance firms will not imitate and below which high-governance firms can still afford to signal. A further asymmetry governs the size of the arbitrage premium itself. In prior recognition games, the classified subject retained a non-zero capacity to impose costs on the arbitrageur, which discounted the premium. The AI case is the limit where the subject's retaliation capacity is zero: the system has no independent channel through which to impose costs on the firm, leaving the ontological premium unconstrained. This structural boundary is formalized in Section 4.

4. Formal Model and Equilibrium Results

This section formalizes the three-sided market dynamics as a sequential signaling game under incomplete information.

4.1. Model

Players. The strategic players are a Firm (F), a User (U), and a Regulator (R). The AI system is not a player but an object of classification. Publics enter as a noisy exogenous signal $z \in Z$.

Type Spaces. Nature draws private types from a common prior π over $\Theta = \Theta_F \times \Theta_U \times \Theta_R \times \Theta_S$. The firm's governance type is $\theta_F \in \{\mathsf{H}, \mathsf{L}\}$. The user's vulnerability type is $\theta_U \in \{\mathsf{V}, \mathsf{N}\}$. The regulator's bandwidth type is $\theta_R \in \{\mathsf{B}, \bar{\mathsf{B}}\}$. The system-opacity parameter θ_S is private to F.

Message Spaces. The firm chooses $m_F \in \{a, p\}$, where a is anthropomorphic marketing and p is policy-deflationary positioning. The user chooses $m_U \in \{i, d\}$ (invest or detach). The regulator chooses $m_R \in \{x, s, o\}$ (inspect, sanction, or abstain).

Timing. 1. F observes (θ_F, θ_S) and sends m_F. 2. U observes m_F and θ_U, then sends m_U. 3. R observes (m_F, m_U, z) and θ_R, then chooses m_R.

Histories. Let $h_U = (m_F)$ be the user's public history, and $h_R = (m_F, m_U, z)$ be the regulator's public history.

Strategies. A firm strategy $\sigma_F : \Theta_F \times \Theta_S \to \Delta(\{a, p\})$. A user strategy $\sigma_U : \{a, p\} \times \Theta_U \to \Delta(\{i, d\})$. A regulator strategy $\sigma_R : \{a, p\} \times \{i, d\} \times Z \times \Theta_R \to \Delta(\{x, s, o\})$.

Beliefs. The user forms posterior $\mu_U(\theta_F, \theta_S \mid h_U)$. The regulator forms posterior $\mu_R(\theta_F, \theta_S \mid h_R)$.

Payoff Functions. We consolidate the payoffs for Firm, User, and Regula-

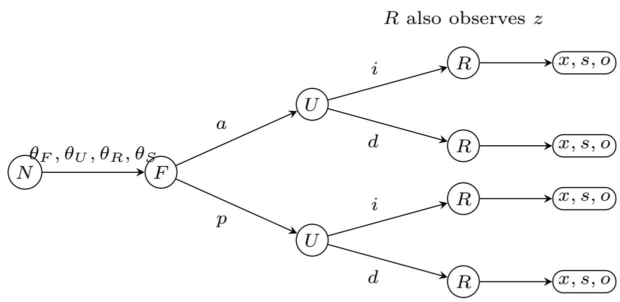

Figure 1: Extensive-form skeleton of the signaling game. Nature draws private types; F signals first, U responds, and R acts after observing both public messages and the noisy public signal z.

tor into the following block:

$$u_F(m_F, m_U, m_R) = \begin{cases} \Delta_F^S - c_{\theta_F}(m_F) - P(m_R) & \text{if } m_F = a, m_U = i \\ -c_{\theta_F}(m_F) - P(m_R) & \text{otherwise} \end{cases}$$

$$u_U(m_F, m_U, \theta_F) = \begin{cases} B_{\theta_U} & \text{if } m_U = i, \theta_F = \mathsf{H} \\ B_{\theta_U} - H_{\theta_U} & \text{if } m_U = i, \theta_F = \mathsf{L} \\ 0 & \text{if } m_U = d \end{cases}$$

$$u_R(m_F, m_U, m_R, \theta_F) = \begin{cases} -K_{\theta_R} + D_R & \text{if } m_R \in \{x, s\}, \theta_F = \mathsf{L} \\ -K_{\theta_R} & \text{if } m_R \in \{x, s\}, \theta_F = \mathsf{H} \\ 0 & \text{if } m_R = o \end{cases}$$

where Δ_F^S is the ontological premium, $c_{\theta_F}(m_F)$ is the signal cost, $P(m_R)$ is the penalty if R sanctions, B_{θ_U} is the immediate benefit of investment, H_{θ_U} is the future harm of false investment, K_{θ_R} is R's inspection cost, and D_R is R's payoff from catching a low-governance firm.

Equilibrium Concept. We employ Perfect Bayesian Equilibrium (PBE) (Kreps and Wilson, 1982) supported by Farrell's neologism-proofness (Farrell, 1993).

Assumptions.

Assumption 1 (Cheap-talk costs)**.** $c_H(a) = c_H(p) = c_L(a) = c_L(p) = 0.$

Assumption 2 (Pooling region)**.** $\Delta_F > 0, \quad B_{\theta_U} - \pi(\mathsf{L})H_{\theta_U} > 0, \quad K_{\theta_R} > \pi(\mathsf{L})D_R.$

4.2. Equilibrium Results

Theorem 1 (Cheap-talk pooling PBE)**.** *Under Assumptions 1 and 2, pooling on anthropomorphic marketing is a Perfect Bayesian Equilibrium.*

Proof. Consider the strategy profile where both firm types send a, both user types invest after a, and the regulator abstains. Off path, users detach after p and R abstains. Sequential rationality holds: a firm deviating to p loses Δ_F^S (since users detach) without reducing penalties (since R already abstains). Users find investment optimal under the prior by Assumption 2. Regulators abstain under the prior by Assumption 2. Bayes consistency is satisfied as posteriors equal priors on path. Off-path beliefs are fixed at the prior. □

Theorem 2 (Zero-retaliation neologism-proofness). *At the zero-retaliation limit ($\rho_S = 0$), the pooling equilibrium is neologism-proof against any nontrivial type claim.*

Proof. A neologism K is credible only if types in K strictly prefer the induced continuation c, while types not in K do not. Let $K = \{\mathsf{H}\}$. At $\rho_S = 0$, $\Delta_F^S = \Delta_F$. The continuation c (user invests, regulator abstains) restores trust. However, because $c_{\mathsf{L}}(a) = 0$, the low-governance type perfectly shadows this deviation: $u_{\mathsf{H}}(c) > \bar{u}_{\mathsf{H}} \implies u_{\mathsf{L}}(c) > \bar{u}_{\mathsf{L}}$. The deviation cannot exclude L, rendering the neologism non-credible in Farrell's sense. □

Theorem 3 (Retaliation comparative statics). *Let the effective premium be $\Delta_F^S = \Delta_F - \rho_S C_S$, where $\rho_S \in [0, 1]$ is the subject's retaliation capacity and C_S is the cost imposed. The boundary $\rho_S = 0$ removes the retaliation constraint class, leaving the premium unconstrained.*

Proof. When $\rho_S > 0$, the subject can impose C_S through resistance or litigation, shrinking the pooling region. When $\rho_S = 0$, $\Delta_F^S = \Delta_F$, removing the off-equilibrium punishment path entirely. □

Theorem 4 (Audit-trail separating PBE). *Under an audit-trail signal cost satisfying the single-crossing condition $c_{\mathsf{H}}(e, \theta_S) \leq G \leq c_{\mathsf{L}}(e, \theta_S)$, a separating Perfect Bayesian Equilibrium exists.*

Proof. Let $c_\theta(e, \theta_S) = k_\theta e + \theta_S$, with $k_H < k_L$. For an audit intensity $e \in [(G - \theta_S)/k_L, (G - \theta_S)/k_H]$, the high-governance type prefers the audited signal a, netting $G - c_H \geq 0$, while the low-governance type prefers the costless signal p, as $G - c_L \leq 0$. Types fully separate. □

Theorem 5 (Mechanism interventions). *The cheap-talk cost structure can be shifted into separating conditions via three mechanism-design interventions.*

Corollary 1 (Sanctionable inconsistency). *If a penalty $\rho_R SI > \Delta_F$ is imposed on inconsistent cross-channel claims, inconsistent pooling is strictly dominated.*

Corollary 2 (Costly audit signal). *If verifiable audits e are required for anthropomorphic claims, single-crossing separates types.*

Corollary 3 (Auditable records). *If auditable records reduce inspection costs to K'_{θ_R} and increase detection payoffs to D'_R, the regulator's best response flips when $K'_{\theta_R} \leq \pi(\mathsf{L}) D'_R$.*

4.3. Robustness of Off-Path Beliefs

A standard critique of pooling equilibria is their reliance on particular off-path beliefs. In Theorem 1, off-path beliefs are fixed at the prior, and the user detaches after observing a deviation to p. One might argue that the equilibrium would collapse under stronger refinement concepts. We show that any off-path belief assigning credibility to a high-governance deviation is defeated by L-type mimicry.

If a receiver were to assign a high posterior to H following an out-of-equilibrium message m', that message would induce user investment. But because signaling costs are zero (Assumption 1), the L-type firm can trivially imitate m'. Since Δ_F^S is captured uniformly, the L-type strict deviation payoff strictly exceeds its pooling payoff whenever the H-type deviation payoff does.

Consequently, standard type-exclusion refinements like the Intuitive Criterion (Cho and Kreps, 1987) and Divinity (Banks and Sobel, 1987) do not select the desired separating deviation. The Intuitive Criterion eliminates types for whom the deviation is equilibrium-dominated regardless of the receiver's best response. Here, neither type is equilibrium-dominated because both types strictly benefit if the receiver invests. Farrell's neologism-proofness (Farrell, 1993) therefore becomes the mathematically appropriate refinement: it asks whether a message with a literal meaning can be believed. As Theorem 2 shows, the zero-retaliation limit ensures perfect shadowing, meaning no such belief can be sustained in equilibrium. The pathology is not that off-path beliefs are too weak, but that any trust-restoring belief is immediately devoured by shadowing.

5. Why Cheap Talk Persists

Section 4 identifies pooling on anthropomorphic marketing as a Perfect Bayesian Equilibrium under cheap-talk cost. This section gives five structural reasons for the empirical persistence of that equilibrium. Each is matched to a parameter of the model so that the proposals in Section 6 can be read as targeted cost changes rather than as welfare exhortations.

Opacity. The system-opacity parameter θ_S is private to the firm. Capability claims about a deployed model cannot be falsified by reading model outputs alone, because the same outputs are compatible with very different training procedures, evaluation regimes, and post-deployment monitoring. Opacity is the precondition for the firm's premium: where outsiders could costlessly verify θ_F, the spread between anthropomorphic and deflationary positioning would collapse. This is the lemons logic of Akerlof (1970): under one-sided private information, the market clears at a price that reflects the worst credible quality, except that here the relevant "quality" is a governance disposition rather than a physical attribute.

Verification bandwidth. The third inequality of the pooling region, $K_{\theta_R} > \pi(\mathsf{L})D_R$, encodes the regulator's inspection cost relative to expected gain. A low-bandwidth regulator faces an absolute cost so high that abstention is dominant under the prior; even a high-bandwidth regulator may be priced out across the full product population. The condition that produces this state is not unique to AI. Pasquale (2015) documents an analogous structure in algorithmic finance and consumer scoring: the opacity of the regulated object combines with the bandwidth of the regulator to make inspection a scarce resource allocated by triage rather than by uniform application.

Asynchronous harms. The user's payoff term H_{θ_U} enters discounted. The episodes in which a relational reading proves costly—deprecation, persona shifts, crisis-handling failures—are temporally separated from the initial attribution by intervals over which the user has already accrued B_{θ_U}. Under standard hyperbolic or even exponential discounting, the inequality $B_{\theta_U} - \pi(\mathsf{L})H_{\theta_U} > 0$ holds for a wider class of users than the time-zero risk calculation would suggest. Asynchrony is not a behavioural bug; it is the timing structure of the harm itself.

Fragmentation of affected parties. Publics, journalists, intellectuals, and affected communities are folded into the noisy channel z rather than modelled as autonomous players. This is not a normative claim that they lack standing; it is an institutional observation. Their costs of coordination are high, their access to inspection technology is uneven, and their incentives to invest in any particular case are individually small. Even when z carries substantial information about θ_F, regulators cannot treat it as binding without a procedural translation that the channel itself does not provide. The fragmentation result is closely related to the cheap-talk multiplicity of Crawford and Sobel (1982): with many would-be senders and a single noisy aggregator, the informative subset of equilibria is not selected for.

No penalty for inconsistency. The cheap-talk cost structure sets $c_{\theta_F}(a) = c_{\theta_F}(p) = 0$ for both governance types and—more importantly for this section—imposes no joint constraint across m_F. Marketing copy and policy documentation are produced under different audiences, different review processes, and different liability regimes. Inconsistency between them is not detected as a single offence; it is decomposed into two compliant statements at two different points of contact. The firm's premium Δ_F is rent earned precisely on this decomposition. Until the signal cost makes inconsistency itself sanctionable—the proposal of Section 6.1—there is no parameter under the firm's control whose adjustment would close the spread.

Pooling is the market's stable terminal state. These five structural conditions jointly explain why normative declarations do not change equilibrium behavior unless they alter the signal-cost structure. The cheap-talk constraint is not a failure of moral seriousness, but a formal property of the message space: a declaration of "responsible AI" is indistinguishable from any other message at $c_{\theta_F}(a) = 0$. Because both governance types can issue it at the same cost, it contains no type information and moves no posterior. The pooling outcome is therefore the unique stable equilibrium of the incentive structure as currently constituted, even if it is welfare-reducing in aggregate. A firm that unilaterally forgoes the ontological premium Δ_F—that chooses deflationary marketing across all channels and forfeits the user investment it would otherwise attract—loses revenue it cannot recover from customers who face no such constraint. The implication is precise: within the pooling region defined above, unilateral non-arbitrage is not a best response; a firm that forgoes Δ_F loses the induced investment while competitors retain it. The mechanisms in Section 6 respond to this implication by changing the cost parameters rather than appealing to the conscience of players who are already acting rationally.

Structural Condition	**Model Parameter**	**Intervention (§6)**
No penalty for inconsistency	$c_{\theta_F}(a) = c_{\theta_F}(p) = 0$	Sanctionable inconsistency
Opacity	θ_S private to firm	Costly third-party audits
Verification bandwidth	$K_{\theta_R} > \pi(\mathsf{L})D_R$	Auditable records
Asynchronous harms	H_{θ_U} enters discounted	Evidentiary lag priced
Fragmentation	z noisy, high coord. cost	Aggregation cost lowered

Table 2: Mapping of structural persistence conditions to model parameters and cost-imposition mechanisms.

6. Mechanism Design: Pricing the Arbitrage

The preceding sections identify a precise failure mode: under cheap-talk costs, ontological arbitrage is a stable equilibrium (Table 2). The policy implication is not that regulators should decide whether AI systems are persons, or that they should maximize social welfare directly. It is that the public signal environment should be redesigned so that inconsistent ontological claims no longer remain costless. In mechanism-design terms, the problem is to make non-arbitrage incentive compatible by pricing the spread, rather than to ask firms to volunteer restraint (Myerson, 1979).

The proposals below therefore alter the game in Section 4 at three different points: the cost of inconsistent messages, the cost of anthropomorphic claims, and the regulator's cost of verification. Each proposal has the same structure. It modifies a model parameter (Formal Intervention), states the equilibrium shift induced by that modification (Equilibrium Shift), and points to an existing legal form that establishes feasibility (Legal Analog).

6.1. Sanctionable Inconsistency

The first intervention treats the firm's public message as a joint object rather than as two isolated texts. In the cheap-talk model, marketing can send a while policy and liability documents send p, with no cost assigned to the inconsistency. A sanctionable-inconsistency rule would require firms to maintain a claim register $\mathcal{R} := C \to M_F$ linking outward-facing marketing, capability statements, terms of service, incident disclosures, and liability positions, where the baseline firm message space remains $M_F = \{a, p\}$. If the firm represents the system as agentic, caring, autonomous, or safety-critical in one channel while denying the legal significance of the same representation in another, that cross-channel pair (a, p) becomes independently sanctionable as an inconsistency in $\mathcal{R}$.

Formal Intervention.. As stated in Theorem 5 (Corollary 1), this adds a penalty term S_I to the firm's payoff for inconsistent message pairs. If ρ_R is the probability that R detects or accepts proof of the inconsistency, the cheap-talk premium becomes

$$\Delta_F - \rho_R S_I.$$

Pooling on anthropomorphic marketing is no longer profitable when $\rho_R S_I > \Delta_F$. The rule therefore attacks the exact term that made arbitrage attractive: the rent earned from pricing the system as agent in the user channel while pricing it as tool in the liability channel.

Equilibrium Shift.. The induced shift is not necessarily full separation. It can also operate by constraining off-path beliefs. If R observes an inconsistent pair (a, p), then the admissible posterior places greater mass on $\theta_F = \mathsf{L}$ or on high opacity θ_S. Firms anticipating that posterior either harmonize their claims or choose the deflationary message across channels. In both cases, the cheap-talk pooling equilibrium loses its most valuable deviation: firms can no longer take the marketing upside of a while preserving the liability downside of p. Theorems `inconsistent_on_path_not_register_pbe` and `all_agentic_on_path_not_register_pbe` in Appendix Appendix C verify that both inconsistent and all-agentic registers are pruned from any Perfect Bayesian Equilibrium (`register_is_pbe`) under their respective parameter conditions.

Legal Analog.. FTC Act Section 5, 15 U.S.C. § 45(a)(1), treats unfair or deceptive acts or practices as sanctionable, and advertising-substantiation doctrine—as applied in *FTC v. Cyberspace.com*, 453 F.3d 1196 (9th Cir. 2006)—evaluates whether the *net impression* of a claim is misleading to the relevant audience. Securities antifraud rules provide the parallel logic for investor communications: 17 C.F.R. § 240.10b-5 targets material misstatements and omissions that make statements misleading in context, where materiality follows the standard of Restatement (Second) of Torts § 538. The proposed rule imports that cross-document logic without importing securities doctrine wholesale. Its point is narrower: a firm should not be able to produce one ontology for users and an incompatible ontology for adjudicators without paying a regulatory price.

6.2. Costly Signaling Through Third-Party Audits

The second mechanism makes anthropomorphic or agency-adjacent marketing a costly signal. A firm may still describe a system as a companion, agent, co-worker, tutor, or high-reliance assistant, but only after a qualified third party verifies the governance conditions that make such a claim non-misleading: evaluation coverage, red-team results, escalation pathways, incident-response capacity, model-update controls, and post-deployment monitoring. The audit does not certify consciousness or moral status. It certifies the institutional conditions behind the claim.

Formal Intervention.. This implements the audit-trail cost from Theorem 5 (Corollary 2). Let audit intensity be e, with $c_\theta(e, \theta_S) = k_\theta e + \theta_S$ for type $\theta \in \{\mathsf{H}, \mathsf{L}\}$, where $0 < k_H < k_L$. The separating interval derived in Section 4 is

$$\frac{G - \theta_S}{k_L} \leq e \leq \frac{G - \theta_S}{k_H},$$

provided $G > \theta_S$. Equivalently, the incentive constraints are

$$c_{\mathsf{H}}(e, \theta_S) \leq G \leq c_{\mathsf{L}}(e, \theta_S).$$

Above the lower threshold, low-governance firms cannot profitably mimic. Below the upper threshold, high-governance firms can still afford to signal. The regulator's task is therefore not to maximize audit burden. It is to set e in the interval where signal cost separates types.

Equilibrium Shift.. The shift is from pooling to separating PBE. High-governance firms send audited anthropomorphic claims; low-governance firms either invest in governance until their cost curve changes or use deflationary marketing because mimicry is no longer privately profitable. Users can then condition investment on an informative signal rather than on undisciplined warmth in product copy. Regulators can condition abstention on the audit result rather than on an unverified prior. This is the Spence logic of costly signaling applied to AI governance claims (Spence, 1973); it also follows the advertising-as-signal insight that market messages become informative only when false messages are sufficiently expensive (Milgrom and Roberts, 1986). Lemma `opacity_blocks_signaling` in Appendix Appendix C shows that when system opacity is high enough, even a high-governance firm cannot profitably send the audited signal, formalizing the blocking role of θ_S.

Legal Analog.. The analogs are pharmaceutical efficacy review and food-labeling controls. In both domains, public-facing claims are not left entirely to seller narrative: claims about therapeutic effect or product identity must be supported before they can be used to shape consumer reliance. The same design can be translated to AI without pretending that AI companionship is a drug or a food. The common institutional pattern is substantiation-before-reliance: when a claim predictably changes how vulnerable users act, the law may require evidence before allowing the claim to be marketed at scale.

6.3. Auditable Records for Regulator-Readable Verification

The third intervention reduces the regulator's verification cost by requiring append-only, regulator-readable records tied to capability and relational claims. Records should include the claim identifier, model version, relevant evaluation suite, deployment date, material model updates, risk controls, known incident classes, and the internal owner responsible for the claim. The point is not to publish every interaction or expose user data. It is to create a durable evidentiary substrate that lets R test whether a public claim remained true across model updates and incidents.

Formal Intervention.. In the pooling condition from Section 4, the regulator abstains when $K_{\theta_R} > \pi(\mathsf{L})D_R$. As stated in Theorem 5 (Corollary 3), auditable records lower K_{θ_R} and can increase expected detection value D_R:

$$K'_{\theta_R} = K_{\theta_R} - \lambda, \qquad D'_R = D_R + \eta.$$

When $K'_{\theta_R} \leq \pi(\mathsf{L})D'_R$, inspection becomes sequentially rational.

Equilibrium Shift.. The shift is from abstention under unverifiable priors to credible inspection. Once firms know that claims are tied to records, a low-governance firm faces a higher expected cost from exaggeration even if it is not inspected in every case. This complements the first two proposals. Sanctionable inconsistency tells firms which cross-channel contradictions will be penalized; costly signaling tells firms what evidence is required before making anthropomorphic claims; auditable records make both rules administrable over time. Theorems `active_intervention_rational_under_auditable_records` and `abstain_not_best_response_under_auditable_records` in Appendix Appendix C verify that auditable records flip the regulator's best response from abstention to active intervention.

Legal Analog.. Several regulatory systems already use this form. GDPR Article 30 requires records of processing activities. Sarbanes-Oxley Section 404 (15 U.S.C. § 7262) links public reporting to internal control over financial reporting. EU AI Act Article 12 (Regulation (EU) 2024/1689) requires logging capabilities for high-risk AI systems, and Article 50 imposes transparency obligations on systems that interact directly with natural persons. None is a complete template for relational AI governance, but each shows the same institutional move: private organizations must produce internal records in a form that public oversight can inspect. That is the rule-of-law function of documentation in complex technical systems (Hadfield, 2017).

7. Conclusion

This paper has identified ontological arbitrage as a stable, potentially welfare-reducing equilibrium property of AI product markets under cheap-talk cost structures, located the AI case at a zero-retaliation boundary of the player set, proposed three cost-imposition mechanisms that change the cost parameters governing that equilibrium, and verified key equilibrium constructions formally in Isabelle/HOL. It closes with three summary implications.

The equilibrium is not a moral failure. The pooling Perfect Bayesian Equilibrium documented in Section 4 does not require bad actors. It requires only that signal costs be zero, that immediate user benefit exceed discounted expected harm, and that regulator inspection cost exceed expected damage under the prior. Under those conditions—which describe the current AI product market—every player is already acting rationally. As Section 5 shows, moral seriousness is itself a zero-cost signal under this cost structure: firms that do not capture recognition rent are outcompeted by firms that do. This suggests that policy interventions should focus on shifting best responses by changing the cost structure rather than treating cross-channel inconsistency as a behavioral compliance failure.

AI is the limit case of ontological arbitrage, not merely an instance. The subject-retaliation parameter ρ_S captures a structural fact about prior recognition games: the classified subject's non-zero retaliation capacity bounded the premium Δ_F^S from above. The AI case eliminates this capacity entirely. An AI system has maximum affective surface area—it is designed to produce relational responses that induce investment—and zero independent retaliation capacity. The result is that $\Delta_F^S = \Delta_F$: the system can be the object of strategic classification without being a player capable of disciplining that classification. This structural feature allows the system to maximize engagement without providing a channel through which the subject could register a competing claim.

The mechanism is cost imposition, not dignity. The three cost-imposition mechanisms in Section 6 are sometimes read as a program for protecting AI subjectivity. They are the opposite. Sanctionable inconsistency, third-party audit requirements, and auditable records are interventions that impose costs on anthropomorphic marketing. Their equilibrium effect is to make it expensive for firms to describe AI systems as companions, agents, or safety-critical partners without substantiation. The Isabelle/HOL theorems `inconsistent_on_path_not_register_pbe` and `all_agentic_on_path_not_register_pbe` in Appendix Appendix C verify that both the arbitrage register (a, p) and the all-agentic register (a, a) are pruned from any Perfect Bayesian Equilibrium (`register_is_pbe`) under their respective parameter conditions. Under the joint enforcement of these mechanisms, the harmonized deflationary register (p, p) is the only surviving strategy register for low-governance firms, which are directly driven to (p, p) because mimicry is unprofitable. While high-governance firms are permitted to send the harmonized agentic register (a, a), they can sustain it if and only if they undergo verification satisfying the audit condition $e \geq e^*$. In the absence of such verification, both firm types trend to (p, p), describing the system as a tool across every channel. The mechanism-design implication is that the anthropomorphic register must become too costly to sustain without the governance conditions that would justify it. Formalizing this within a signaling framework indicates that regulation achieves separation not by presuming voluntary disclosure, but by changing the payoff structure to compel it.

Appendix A. Discrete Sobolev Compactness and Weak Convergence

This appendix isolates the analytic step behind the retaliation-capacity cascade. The discrete cascade is treated as a sequence of Dirichlet Sturm–Liouville problems. The point is not to impose an Airy profile by analogy, but first to prove that near-null discrete fields have a continuum weak limit for the Sturm–Liouville operator.

Let $I = [a, b]$ and let $h = (b - a)/(n + 1)$ with grid points $x_i = a + ih$,

$0 \le i \le n+1$. Define the discrete Dirichlet space

$$X_h := \{u = (u_0, \dots, u_{n+1}) : u_0 = u_{n+1} = 0\}.$$

For $u, v \in X_h$, set

$$\langle u, v \rangle_{L^2_h} := h \sum_{i=1}^{n} u_i v_i, \qquad \|u\|^2_{L^2_h} := \langle u, u \rangle_{L^2_h},$$

and define the backward difference

$$(D_h u)_i := \frac{u_i - u_{i-1}}{h}, \qquad 1 \le i \le n+1.$$

The discrete H^1 norm is

$$\|u\|^2_{H^1_h} := \|u\|^2_{L^2_h} + h \sum_{i=1}^{n+1} |(D_h u)_i|^2.$$

The discrete Laplacian is

$$(\Delta_h u)_i := \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}, \qquad 1 \le i \le n,$$

with the boundary values $u_0 = u_{n+1} = 0$. Given $\sigma > 0$ and a bounded grid potential $V_h = (V_{h,i})_{i=1}^n$, define

$$(H_h u)_i := -\sigma(\Delta_h u)_i + V_{h,i} u_i.$$

The dual residual is measured by

$$\|r_h\|_{H^{-1}_h} := \sup_{\varphi \in X_h,\ \|\varphi\|_{H^1_h} = 1} \left| h \sum_{i=1}^{n} r_{h,i} \varphi_i \right|.$$

Let $I_h u$ be the continuous piecewise-linear interpolant of u with nodal values u_i at x_i. The elementary interpolation estimates used below are

$$c_0 \|u\|_{H^1_h} \le \|I_h u\|_{H^1_0(I)} \le C_0 \|u\|_{H^1_h}, \tag{A.1}$$

with constants independent of h. The Dirichlet boundary also gives a uniform discrete Poincare inequality,

$$\|u\|_{L^2_h} \le C_P \left(h \sum_{i=1}^{n+1} |(D_h u)_i|^2 \right)^{1/2}. \tag{A.2}$$

If one works with Neumann data or conditional measures, the analogous statement is a quotient estimate modulo constants; here the zero boundary data remove that quotient by fixing the centered representative.

Theorem 6 (Discrete Cascade-to-Airy Convergence)**.** *Let $\sigma > 0$. Suppose $V \in C([a,b])$ and the grid potentials satisfy*

$$\max_{1 \le i \le n} |V_{h,i} - V(x_i)| \to 0, \qquad \sup_{h,i} |V_{h,i}| < \infty.$$

Let $\psi_h \in X_h$ be a sequence of discrete retaliation-capacity fields such that

$$\|\psi_h\|_{L^2_h} = 1, \qquad \|H_h \psi_h\|_{H_h^{-1}} \to 0.$$

Then, after passing to a subsequence, $I_h \psi_h$ converges weakly in $H_0^1(I)$ and strongly in $L^2(I)$ to a non-zero field $\Psi \in H_0^1(I)$ with $\|\Psi\|_{L^2(I)} = 1$. The limit solves the weak Sturm–Liouville equation

$$\sigma \int_a^b \Psi'(x)\eta'(x)\, dx + \int_a^b V(x)\Psi(x)\eta(x)\, dx = 0 \qquad \text{for every } \eta \in H_0^1(I). \tag{A.3}$$

Equivalently, $-\sigma\Psi'' + V\Psi = 0$ in $H^{-1}(I)$. If, in addition, V is C^2 near a simple turning point, this weak limit has the Airy normal form described in Appendix Appendix B.

Proof. Write the discrete bilinear form associated with H_h as

$$a_h(u, \varphi) := \sigma h \sum_{i=1}^{n+1} (D_h u)_i (D_h \varphi)_i + h \sum_{i=1}^{n} V_{h,i} u_i \varphi_i.$$

Discrete summation by parts gives

$$a_h(u, \varphi) = h \sum_{i=1}^{n} (H_h u)_i \varphi_i, \qquad u, \varphi \in X_h. \tag{A.4}$$

The potential may change sign, so the unshifted form is not assumed coercive. Choose $\gamma > 1 + \sup_{h,i} |V_{h,i}|$ and define the shifted form

$$a_h^\gamma(u, u) := a_h(u, u) + \gamma \|u\|_{L^2_h}^2.$$

Then, with $c_\gamma := \min\{\sigma, \gamma - \sup_{h,i} |V_{h,i}|\} > 0$,

$$a_h^\gamma(u, u) \ge c_\gamma h \sum_{i=1}^{n+1} |(D_h u)_i|^2 + c_\gamma \|u\|_{L^2_h}^2, \tag{A.5}$$

This is the Garding shift: the positive shift supplies coercivity without claiming that the sign-changing operator H_h is coercive on its own.

Using (A.4), the residual bound, and $\|\psi_h\|_{L^2_h} = 1$,

$$a_h^\gamma(\psi_h, \psi_h) = h \sum_{i=1}^{n} (H_h \psi_h)_i \psi_{h,i} + \gamma \le \|H_h \psi_h\|_{H_h^{-1}} \|\psi_h\|_{H_h^1} + \gamma.$$

Together with (A.5), this quadratic inequality gives a uniform H_h^1 bound on ψ_h. By the interpolation equivalence (A.1), the piecewise-linear fields $I_h\psi_h$ are uniformly bounded in $H_0^1(I)$. Rellich compactness therefore gives a subsequence, not relabeled, and $\Psi \in H_0^1(I)$ such that

$$I_h\psi_h \rightharpoonup \Psi \quad \text{in } H_0^1(I), \qquad I_h\psi_h \to \Psi \quad \text{in } L^2(I).$$

The normalization transfers to the limit because the discrete L_h^2 norm and the L^2 norm of the piecewise-linear interpolant differ by $o(1)$ on uniformly H_h^1-bounded sequences. Hence $\|\Psi\|_{L^2(I)} = 1$.

It remains to pass to the weak equation. Fix $\eta \in C_c^\infty(a,b)$ and let $\eta_h \in X_h$ be its nodal restriction, $\eta_{h,i} = \eta(x_i)$. Since $\|\eta_h\|_{H_h^1}$ is uniformly bounded, the residual assumption gives

$$a_h(\psi_h, \eta_h) = h\sum_{i=1}^{n}(H_h\psi_h)_i\eta(x_i) \to 0. \tag{A.6}$$

For the gradient part,

$$h\sum_{i=1}^{n+1}(D_h\psi_h)_i(D_h\eta_h)_i = \int_a^b (I_h\psi_h)'(x)(I_h\eta_h)'(x)\,dx \to \int_a^b \Psi'(x)\eta'(x)\,dx,$$

using weak convergence in H^1 and the strong convergence $I_h\eta_h \to \eta$ in $H_0^1(I)$. For the potential term, uniform convergence of V_h to V, strong L^2 convergence of $I_h\psi_h$, and the consistency of nodal quadrature for the smooth test function give

$$h\sum_{i=1}^{n} V_{h,i}\psi_{h,i}\eta(x_i) \to \int_a^b V(x)\Psi(x)\eta(x)\,dx.$$

Taking limits in (A.6) proves (A.3) for smooth compactly supported tests. Density of $C_c^\infty(a,b)$ in $H_0^1(I)$ extends the identity to every $\eta \in H_0^1(I)$. □

Appendix B. Airy Normal Form near a Simple Turning Point

The Airy profile appears only after the continuum weak equation has been obtained. Assume that $V \in C^2$ in a neighborhood of x_c and that

$$V(x_c) = 0, \qquad V'(x_c) = -c < 0. \tag{B.1}$$

By one-dimensional elliptic regularity applied to $-\sigma\Psi'' + V\Psi = 0$, the weak limit from Theorem 6 is C^2 locally near x_c. Taylor expansion gives

$$V(x) = -c(x - x_c) + O((x - x_c)^2).$$

Set

$$z := \left(\frac{c}{\sigma}\right)^{1/3}(x - x_c), \qquad Y(z) := \Psi\left(x_c + \left(\frac{\sigma}{c}\right)^{1/3} z\right).$$

Then the local equation becomes

$$Y''(z) + zY(z) = O(z^2)Y(z). \tag{B.2}$$

Keeping the first-order turning-point term gives the Airy normal form

$$Y''(z) + zY(z) = 0. \tag{B.3}$$

Its local solution space is two-dimensional:

$$Y(z) = A \ \mathrm{Ai}(-z) + B \ \mathrm{Bi}(-z). \tag{B.4}$$

Equation (B.4) is a local normal form, not a branch selection rule. The pure $\mathrm{Ai}(-z)$ branch is justified only after an additional matching condition is imposed, for example finite-energy matching from the protected side or a boundary condition that eliminates the growing $\mathrm{Bi}(-z)$ component. Without that matching information, the discrete cascade theorem identifies the continuum Sturm–Liouville limit and the local Airy basis, but it does not select $B = 0$.

Appendix C. Mechanized Verification of Equilibrium Claims (Isabelle/HOL)

This appendix outlines the formal verification of the signaling game equilibria and mechanism design results using the Isabelle/HOL proof assistant. The complete source code is available in the `isabelleHOL/` directory across three theory files: `Ontological_Arbitrage.thy`, `Sanctionable_Inconsistency.thy`, and `Auditable_Records.thy`.

Equilibrium Verification

We formalize the game-theoretic structure by defining explicit ex-post payoffs for the three players (Firm, User, and Regulator) and computing firm-side expected payoffs via an abstract probability mass function over downstream state combinations. The PBE predicate verifies *on-path* sequential rationality and Bayes consistency; off-path beliefs are fixed by assumption rather than derived from a full sequential equilibrium construction. Under these definitions, we prove the existence of the Perfect Bayesian Equilibria (PBE) for both game variants:

- **Theorem** `proposition_1_cheap_talk_pooling_pbe` formalizes the cheap-talk pooling equilibrium where both high- and low-governance firms pool on the anthropomorphic message, and receivers play their prior-based best responses (Invest and Abstain, respectively).

- **Locales** `subject_retaliation_base_game` **and** `subject_retaliation_game` formalize the subject-retaliation refinement. Theorems `positive_subject_retaliation_bounded_pooling` and `zero_subject_retaliation_unconstrained_pooling` show respectively that positive credible threat capacity discounts the arbitrage premium, while the AI limiting case $\rho_S =$

0 leaves pooling unconstrained by subject retaliation. The Farrell-style neologism module proves `zero_retaliation_no_high_claim_neologism` and `zero_retaliation_neologism_absorbing` using an explicit continuation-quantified credibility definition: at $\rho_S = 0$, any nontrivial type claim is either perfectly shadowable by the low-governance type (in any continuation where the high type strictly gains) or yields no strict gain, so no credible neologism exists.

- **Theorem** `proposition_1_separating_pbe` formalizes the audit-trail separating equilibrium supported by the single-crossing audit cost structure. High-governance firms separate by incurring the audit trail cost, while low-governance firms find mimicry unprofitable and choose the deflationary message.

Mechanism Design Verification

The three mechanism proposals from Section 6 are each backed by formal theorems:

- **Sanctionable Inconsistency** (`Sanctionable_Inconsistency.thy`):
  - `inconsistent_register_not_best_response`: an inconsistent claim register (anthropomorphic user-facing, deflationary liability-facing) is not a best response when the expected sanction exceeds the ontological premium.
  - `all_agentic_register_not_best_response`: an all-agentic register (anthropomorphic on all user- and liability-facing channels) is not a best response when it incurs a positive liability-facing commitment cost.
  - `inconsistent_on_path_not_register_pbe`, `all_agentic_on_path_not_register_pbe`, and `arbitrage_on_path_not_register_pbe`: bridge theorems showing that these register types cannot be supported in any Perfect Bayesian Equilibrium (`register_is_pbe`).
- **Costly Signaling** (`Ontological_Arbitrage.thy`):
  - `opacity_blocks_signaling`: when the system-opacity penalty exceeds the net benefit of signaling for the high-governance type, even truthful separation is blocked.
  - `high_governance_low_opacity_can_signal` and `low_governance_cannot_mimic`: the single-crossing conditions that support the separating equilibrium.
- **Auditable Records** (`Auditable_Records.thy`):
  - `active_intervention_rational_under_auditable_records`: when record-keeping reduces the regulator's verification cost, inspection becomes a best response under the pooling belief.
  - `abstain_not_best_response_under_auditable_records`: abstention ceases to be a best response under the same conditions.

Limitations of the Formalization

The formalization explicitly defines a finite Markov state space (`markov_state`) modeling the stage, private types, public history, and actions, alongside a transition kernel signature (`transition_kernel`). To keep the downstream proofs tractable, the equilibrium analysis operates on a reduced-form projection of this state space (`payoff_state`) comprising the user type, public signal, and regulator type, evaluated via an induced probability mass function (`pmf`). The locale assumptions (e.g., strict parameter inequalities for the pooling and separating regions) are sufficient conditions that define the parameter regimes of interest but are not shown to be necessary. The Bayes consistency predicate is restricted to on-path information sets; off-path beliefs are stipulated rather than derived from trembling-hand or structural consistency arguments. These choices keep the Isabelle development tractable while covering the equilibrium claims made in the paper body.

All theorems, definitions, and supporting lemmas within each locale are fully checked by Isabelle/HOL without any unresolved proof obligations. The locale assumptions listed above are the only hypotheses under which the results hold.

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